

Homework Assignment: Principal PM – Connected Asset Maintenance

Background

Samsara's Connected Asset Maintenance product helps fleet and equipment operators manage work orders, preventive maintenance schedules, parts inventory, and vendor relationships — all connected to the real-time vehicle and equipment data flowing through Samsara's IoT platform (fault codes, engine diagnostics, usage data, GPS location, and more).

Today, a maintenance supervisor starting their morning shift has to piece together “what should my team work on today?” from multiple sources — work order lists, overdue PM schedules, DVIR defect reports, active fault codes, and their own mental model of what's in the shop. Most of this planning still happens on whiteboards and in people's heads, even when they have software tools available.

We believe there's an opportunity to build an experience that helps maintenance supervisors plan their team's daily work — bringing together everything that needs attention, who's available to work on it, and what should be prioritized.

The Assignment

Part 1: Product Memo

Write a short product memo (strict 4 page maximum, shorter is fine) describing how you'd approach this problem. Include:

- The value this will provide to customers and why it matters
- How you'd measure success
- Roughly what you think the UX should be
- A clear recommendation on where and how to start (what's in v1 vs. what comes later)

Use Samsara's website (samsara.com/products/maintenance) to familiarize yourself with the current product. Research how fleet maintenance operations actually work — the more specific and grounded your thinking, the better. Make assumptions as necessary (we won't be answering questions about the assignment) and state them where relevant. Include a set of open questions you'd want to dig into further.

The strongest memos have a clear, focused vision and a specific point of view on what matters most. We value concise, direct, plain-language writing.

Use of AI tools for research and writing is highly encouraged, though you must thoroughly review and stand by the final proposal.

Part 2: Working Prototype

Using AI coding tools (Cursor, Claude Code, Codex, or any AI-assisted coding tool of your choice), build a working prototype of the core experience you described in your memo. This should be a real application with a frontend and a backend — not a Figma mockup or a static page. Mocking input data that would likely be present in Samsara’s system is of course fine.

The prototype doesn’t need to be production-ready, but it should be functional enough to demonstrate the key interactions and give us a feel for how the experience would work.

We’re evaluating:

- **Product thinking** — Does the prototype reflect a clear, opinionated view of what matters to the user?
- **Research depth** — Does the experience reflect thorough domain research and genuine curiosity about how maintenance operations actually run?
- **Craft and ambition** — How far did you push it? Did you consider the nuances of the UX, even in an early prototype?
- **AI tool fluency** — How effectively did you leverage AI tools to move fast? We’d love to hear briefly about your process.

You’re welcome to use any tech stack. Include a README with instructions to run the app locally.

Submission

Please send both the memo (PDF or Google Doc) and a link to the prototype (providing the files and a README to run locally is fine — no need to host it).